Solution requirements

Date	30 September 2026
Team id	SWTID-2026-2551
Project name	Fit Buddy – AI Fitness Plan Generator using Gemini Models
Maximum mark	4 marks

Step 1: Functional Requirements (FR)

S.No	**Requirement Category**	**Requirement Description**	**Priority**
1	Authentication	Users can enter their basic details to access the FitBuddy fitness planning service.	High
2	Authorization levels	User can generate and update fitness plans. Admin can view registered users and their plans.	Medium
3	External interfaces	The system integrates with Google Gemini AI models for workout plan and nutrition tip generation.	High
4	Transactions processing	The system receives user fitness information, processes it through the AI model, and returns a personalized fitness plan.	High
5	Reporting	Admin can view user details, original workout plans, and updated workout plans.	Medium
6	Business rules	The system generates a personalized 7-day workout plan based on	High

		age, weight, fitness goal, and workout intensity.	
7	Compliance to laws or regulations	User information should be handled securely and privacy should be maintained.	Medium
8	Other	Users can provide feedback and update their existing workout plan using AI.	High

Step 2: Non-Functional Requirements (NFR)

Group No.	Final Idea	Feasibility	Importance	Priority	Selected
1	AI-based personalized 7-day workout plan using Gemini	High	High	High	Yes
2	AI-generated nutrition and recovery tips	High	High	Medium	Yes
3	User data storage and easy web interface	High	Medium	Medium	Yes

Main Solution:

Fit Buddy uses **Fast API, Google Gemini AI, Jinja2, SQL Alchemy, and SQLite** to generate personalized fitness plans and update them based on user feedback.